Sequences — problem 9.36

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The sequence a_n is defined as follows : $a_{4n+1} = 1$, $a_{4n+3} = 0$ for $n \geq 0$, and $a_{2n} = a_n$ for $n \geq 1$. Show that this sequence is not periodic.

Remark. These digits can be used to draw a curve as follows : Start at the origin and go one step to the right. If the next bit is 1, then turn left by 90° and go one step forward. If the next bit is 0 turn right by 90° and go one step forward. You get a strange curve with many regularities, which is called a "dragon curve."